

Feedback stabilization of delayed semilinear discrete systems

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 - Motivation
 - Definitions and assumptions

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Introduction

Motivation

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 - more flexible than linear systems,
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Example of interest

$$\begin{cases} x(k+1) = Ax(k) + v(k)Bx(k-h) & k \in \mathbb{N} \\ x(k) = \Psi(k) & k \in \llbracket -h, 0 \rrbracket \end{cases}$$

Problem statement

Or more general

$$\begin{cases} x(k+1) = Ax(k) + v(k)B(x_k), & k \in \mathbb{N} \\ x_0 = \varphi \in \mathcal{C}, \end{cases} \quad (1)$$

Problem statement

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$$\begin{cases} x(k+1) = Ax(k) + v(k)B(x_k), & k \in \mathbb{N} \\ x_0 = \varphi \in \mathcal{C}, \end{cases} \quad (1)$$

where

- $\mathcal{H}$ is a Hilbert space,
- $A \in \mathcal{L}(\mathcal{H})$ with $\|A\| \leq 1$,
- $B : \mathcal{C} \mapsto \mathcal{H}$ is locally Lipschitz and $B(0) = 0$,
- $k \mapsto v(k)$ represents the control,
- $x_k \in \mathcal{C} := \{\text{Set of all mappings } \varphi : \llbracket -h, 0 \rrbracket \longrightarrow \mathcal{H} \mid (h \geq 0) \text{ equipped with norm } \|\varphi\|_{\mathcal{C}} := \max_{\theta \in \llbracket -h, 0 \rrbracket} \|\varphi(\theta)\|\}$, is defined as follows:

$$x_k(\theta) = x(k + \theta), \quad \forall \theta \in \llbracket -h, 0 \rrbracket,$$

- φ is a given function.

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From continuous to discrete

System (1) can also arise from the semi-Euler discretization of the following delayed continuous-time system:

$$\begin{cases} \dot{x}(t) = A_c x(t) + u_c(t) B_c x(t-h), & t \geq 0 \\ x(t) = \Psi(t), & t \in [-h, 0] \end{cases} \quad (2)$$

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where

- A_c is a linear operator generating a strongly continuous semi-group of contractions on $\mathcal{H}$.
- $u_c(t)$ is a real valued control.

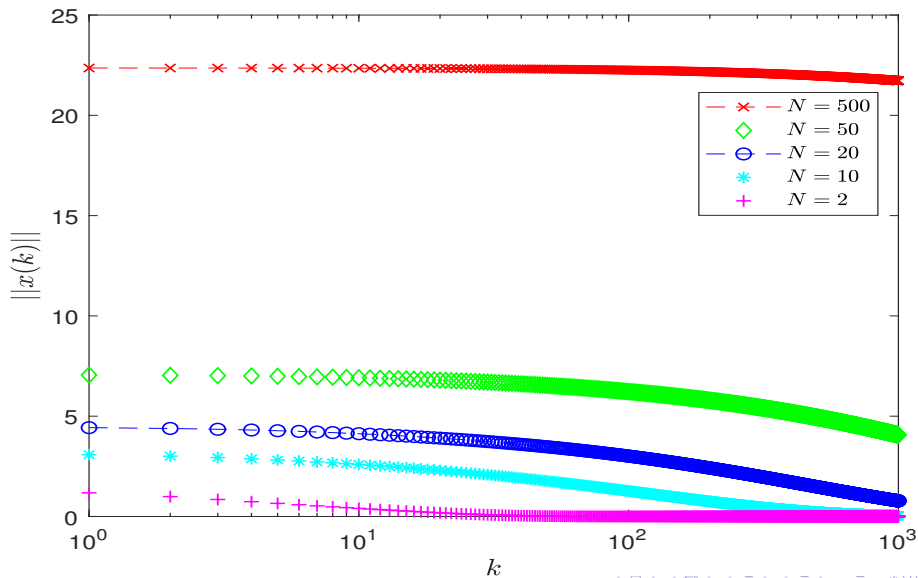
In system (1), the operators become: $A = R_\lambda = (I - \lambda A_c)^{-1}$, $B = \lambda R_\lambda B_c$, where $\lambda \in (0, +\infty)$, and $\|A\| \leq 1$.

- Larbi Berrahmoune. “Stabilization and decay estimate for distributed bilinear systems”. In: *Systems & control letters* 36.3 (1999)
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High Dimensionality Problem



Failure of the Lifting method

Taking $B \in \mathcal{L}(\mathcal{H})$ and $h = 1$ in system (1) and using the 'Lifting method', we get

$$X(k) := \begin{pmatrix} x(k) \\ x(k-1) \end{pmatrix}, \quad X(0) = \begin{pmatrix} \psi(0) \\ \psi(-1) \end{pmatrix}, \quad A_{\text{aug}} = \begin{pmatrix} A & 0 \\ I & 0 \end{pmatrix}$$

and $B_{\text{aug}} = \begin{pmatrix} 0 & B \\ 0 & 0 \end{pmatrix}.$

System (1) becomes:

$$X(k+1) = A_{\text{aug}}X(k) + u(k)B_{\text{aug}}X(k), \quad \forall k \in \mathbb{N}. \quad (3)$$

- Sufficient conditions proposed in ([1, 5, 9]) do not hold for the system (3).

Definitions

Definition 1 :

The equilibrium at the origin (i.e $\bar{x} = 0$) for system (1) is called stable, if for every $\varepsilon > 0$, there exists $\delta > 0$ such that $\|\varphi\|_{\mathcal{C}} < \delta \implies \|x(k)\| < \varepsilon, \forall k \geq 0$.

Definition 2 :

System (1) is called strongly stable if the origin is a stable equilibrium point of system (1) and for all $\varphi \in \mathcal{C}$, we have $\|x(k)\| \rightarrow 0$ as $k \rightarrow +\infty$.

Definitions

Definition 3 :

System (1) is called polynomially stable, if the origin is a stable equilibrium point of system (1) and there exists $\alpha > 0$ such that

$$\|x(k)\| = \mathcal{O}\left(\frac{1}{k^\alpha}\right), \text{ as } k \rightarrow +\infty.$$

Definition 4 :

- ❶ System (1) is called exponentially stable, if the origin is a stable equilibrium point of system (1) and there exist two constants $P \geq 1$ and $q > 0$ such that

$$\|x(k)\| \leq P e^{-qk} \|\varphi\|_{\mathcal{E}}, \quad \forall k \in \mathbb{N}. \quad (4)$$

- ❷ If P and q are independent of φ , system (1) is said to be uniformly exponentially stable.

Assumptions

Let

$$v(k) = -\rho \frac{\langle Ax(k), B(x_k) \rangle}{\|x_k\|_{\mathcal{C}}^2} 1_E(k), \quad k \in \mathbb{N}, \quad (5)$$

where $1_E(k)$ is the characteristic function of the set $E := \{k \in \mathbb{N} : \|x_k\|_{\mathcal{C}} \neq 0\}$ and $\rho > 0$.

($\mathcal{A}_1$) $\exists \delta > 0, T \geq h$, for all $y \in \mathcal{H}$, we have

$$\sum_{k=h}^T \left| \left\langle B(A^{k+1} y), A^{k+1} y \right\rangle \right| \geq \delta \|y\|^r,$$

where $r \geq 2$ is a real.

For $\eta \in (0, \infty]$, we define $\Omega(\eta) := \{\varphi \in \mathcal{C} : \|\varphi\| \leq \eta\}$ and let L_η be the Lipschitz constant of B .

Remark 1 :

The feedback control (5) ensures that the energy of system (1) is nonincreasing.

Indeed, for $k \in \mathbb{N}$, we have

$$\|x(k+1)\|^2 - \|x(k)\|^2 \leq \rho (\rho L_\eta^2 - 2) \frac{|\langle Ax(k), B(x_k) \rangle|^2}{\|x_k\|_{\mathcal{C}}^2}. \quad (6)$$

Then, for $\rho \in \left(0, \frac{2}{L_\eta^2}\right)$, the sequence $(\|x(k)\|)_{k \geq 0}$ is nonincreasing.

Remark 2 :

If $B \in \mathcal{L}(\mathcal{H})$, $L_\eta = \|B\|$.

Main results

Exponential Stabilization

Theorem 1

Assume that $(\mathcal{A}_1)$ is satisfied for $r = 2$. Then, for all $\varphi \in \Omega(\eta)$ and all $\rho \in \left(0, \frac{2}{L_\eta^2}\right)$, the feedback control (5) exponentially stabilizes system (1).

Proof:

Using the variation of constants formula, the solution of system (1) is given by:

$$\begin{cases} x(k) = A^k x(0) + \sum_{s=0}^{k-1} A^{k-1-s} v(s) B(x_s), & k \in \mathbb{N}^* \\ x_0 = \varphi \in \mathcal{C}. \end{cases} \quad (7)$$

Proof

From (7), we have

$$\begin{aligned} |\langle B(A^{k+\cdot}x(0)), A^{k+1}x(0) \rangle| &\leq 2L_\eta^2 \sqrt{T} \sqrt{\sum_{s=0}^{T-1} |v(s)|^2 \|x_s\|_\mathcal{C}^2} \|x(0)\| \\ &+ \frac{1}{\rho} |v(k)| \|x_k\|_\mathcal{C} \|x(0)\|. \end{aligned}$$

Summing the last inequality from h to T , we get

$$\begin{aligned} \sum_{k=h}^T |\langle B(A^{k+\cdot}x(0)), A^{k+1}x(0) \rangle| &\leq \left(2L_\eta^2 (T+1) \sqrt{T} \sqrt{\sum_{s=0}^{T-1} |v(s)|^2 \|x_s\|_\mathcal{C}^2} \right. \\ &\quad \left. + \frac{1}{\rho} \sqrt{T} \sqrt{\sum_{s=0}^T |v(s)|^2 \|x_s\|_\mathcal{C}^2} \right) \|x(0)\|. \end{aligned}$$

Proof

Using $(\mathcal{A}_1)$ and replacing $x(0)$ with $x(m)$, where $m \in \mathbb{N}$, we obtain

$$\frac{\delta^2 \|x(m)\|^4}{\left(2L_\eta^2(T+1)\sqrt{T} + \frac{1}{\rho}\sqrt{T}\right)^2} \leq \sum_{s=m}^{m+T} |v(s)|^2 \|x_s\|_{\mathcal{C}}^2 \|x(m)\|^2.$$

Recall that

$$\|x(k+1)\|^2 - \|x(k)\|^2 \leq \rho(\rho L_\eta^2 - 2) \frac{|\langle Ax(k), B(x_k) \rangle|^2}{\|x_k\|_{\mathcal{C}}^2}.$$

Combining the two last inequalities, we get

$$\frac{\delta^2 \|x(m)\|^4}{\left(2L_\eta^2(T+1)\sqrt{T} + \frac{1}{\rho}\sqrt{T}\right)^2} \leq \sum_{s=m}^{m+T} |v(s)|^2 \|x_s\|_{\mathcal{C}}^2 \|x(m)\|^2.$$

Proof

For $m = n(T + 1)$, with $n \in \mathbb{N}$, we get

$$\|x(n+1)(T+1)\|^2 \leq \left(1 + \frac{\delta^2 \left(\frac{2}{\rho} - L_\eta^2 \right)}{\left(2L_\eta^2(T+1)\sqrt{T} + \frac{1}{\rho}\sqrt{T} \right)^2} \right)^{-1} \|x(n(T+1))\|^2.$$

By induction, we obtain

$$\|x(k)\| \leq M(\rho)e^{-q(\rho)k} \|\varphi\|_{\mathcal{C}}, \quad \forall k \in \mathbb{N}, \quad (8)$$

where $M(\rho) := c^{-\frac{1}{2}}(\rho)$ and $q(\rho) := -\frac{\ln(c(\rho))}{2(T+1)} > 0$, with

$$c(\rho) := \left(1 + \frac{\delta^2 \left(\frac{2}{\rho} - L_\eta^2 \right)}{\left(2L_\eta^2(T+1)\sqrt{T} + \frac{1}{\rho}\sqrt{T} \right)^2} \right)^{-1} \in (0, 1).$$

Polynomial Stabilization

Theorem 2 :

Assume that $(\mathcal{A}_1)$ is satisfied for $r > 2$. Then, for all $\varphi \in \Omega(\eta)$ and all $\rho \in \left(0, \frac{2}{L_\eta^2}\right)$, the feedback control (5) strongly stabilizes system (1) with the following decay rate:

$$\|x(k)\| = \mathcal{O}\left(k^{-\frac{1}{2(r-1)}}\right), \text{ as } k \rightarrow +\infty.$$

First, we will need the following Lemma

Lemma 1 [[2]]

Let (α_k) be a sequence of positive real numbers satisfying

$$\alpha_{k+1} \leq \alpha_k - R\alpha_{k+1}^{2+m}, \quad \forall k \geq 0,$$

where $R > 0$ and $m > -1$ are constants. Then, there exists a positive constant D (depending on m and R) such that

$$\alpha_k \leq \frac{D}{(k+1)^{\frac{1}{1+m}}}, \quad \forall k \geq 0.$$

Proof

Following the same approach as in the previous proof, we have

$$\begin{aligned} \sum_{k=h}^T |\langle B(A^{k+1}x(0)), A^{k+1}x(0) \rangle| &\leq \left(2L_\eta^2(T+1)\sqrt{T} \sqrt{\sum_{s=0}^{T-1} |v(s)|^2 \|x_s\|_{\mathcal{C}}^2} \right. \\ &\quad \left. + \frac{1}{\rho} \sqrt{T} \sqrt{\sum_{s=0}^T |v(s)|^2 \|x_s\|_{\mathcal{C}}^2} \right) \|x(0)\| \end{aligned}$$

and

$$\|x(k+1)\|^2 - \|x(k)\|^2 \leq \rho(\rho L_\eta^2 - 2) \frac{|\langle Ax(k), B(x_k) \rangle|^2}{\|x_k\|_{\mathcal{C}}^2}.$$

Proof

Using $(\mathcal{A}_1)$, for $r > 2$, we obtain

$$\begin{aligned} \|x((n+1)(T+1))\|^2 - \|x(n(T+1))\|^2 &\leq \left(L_\eta^2 - \frac{2}{\rho}\right) \sum_{k=n(T+1)}^{n(T+1)+T} |v(k)|^2 \|x_k\|_{\mathcal{C}}^2 \\ &\leq \frac{\delta^2 \left(L_\eta^2 - \frac{2}{\rho}\right) \|x(n(T+1))\|^{2(r-1)}}{\left(2L_\eta^2(T+1)\sqrt{T} + \frac{1}{\rho}\sqrt{T}\right)^2}. \end{aligned}$$

Setting $\alpha_k = \|x(k(T+1))\|^2$, we infer that

$$\alpha_{k+1} \leq \alpha_k - R\alpha_{k+1}^{r-1},$$

$$\text{where } R := \frac{\delta^2 \left(\frac{2}{\rho} - L_\eta^2\right)}{\left(2L_\eta^2(T+1)\sqrt{T} + \frac{1}{\rho}\sqrt{T}\right)^2} > 0.$$

Using **Lemma 1**, we get the desired estimation

$$\|x(k)\| = \mathcal{O}\left(k^{-\frac{1}{2(r-1)}}\right), \text{ as } k \rightarrow +\infty. \quad (9)$$

Bilinear case

Finite dimensional case

Suppose that $B \in \mathcal{L}(\mathcal{H})$ and $r = 2$.

Proposition 1 :

If $\dim(\mathcal{H}) < \infty$, then $(\mathcal{A}_1)$ is equivalent to

$(\mathcal{A}_2) \quad \forall y \in \mathcal{H}$, if $\langle BA^{k-h}y, A^{k+1}y \rangle = 0, \forall k \geq h$, then $y = 0$.

Proof:

- $\Rightarrow$ One can clearly see that $(\mathcal{A}_1)$ implies $(\mathcal{A}_2)$.
- $\Leftarrow$ Let $y \neq 0$ and $T \geq h$, we define

$$\delta = \min_{\|y\|=1} \sum_{k=h}^T |\langle B_u A_u^{k-h} y, A_u^{k+1} y \rangle|. \quad (10)$$

Proceeding by contradiction and using the continuity of (10) on the unit sphere, we get $\bar{y} = 0$, where $y_{T_{\varphi(n)}} \rightarrow \bar{y}$. This contradicts the fact that $\|\bar{y}\| = 1$.

Decomposition approach

- $\mathcal{H} = \mathcal{H}_u \overset{\perp}{\oplus} \mathcal{H}_s,$

where $\mathcal{H}_u$ and $\mathcal{H}_s$ are two closed subspaces of $\mathcal{H}$, which are orthogonal and invariant under A and B . System (1) can be divided into the following subsystems :

$$\begin{cases} x_u(k+1) = A_u x_u(k) + v_u(k) B_u x_u(k-h), & k \in \mathbb{N} \\ x_u(k) = \psi_u(k), & k \in \llbracket -h, 0 \rrbracket, \end{cases} \quad (11)$$

and

$$\begin{cases} x_s(k+1) = A_s x_s(k) + v_u(k) B_s x_s(k-h), & k \in \mathbb{N} \\ x_s(k) = \psi_s(k), & k \in \llbracket -h, 0 \rrbracket. \end{cases} \quad (12)$$

Generally, system (12) without control is exponentially stable and $\dim(\mathcal{H}_u) < +\infty$.

Remark 3 :

($\mathcal{A}_1$) with $\mathcal{H} = \mathcal{H}_u$ is equivalent to:

($\mathcal{A}_2$) $\forall y \in \mathcal{H}_u$, if $\langle B_u A_u^{k-h} y, A_u^{k+1} y \rangle = 0, \forall k \geq h$, then $y = 0$.

Theorem 3 :

Suppose that ($\mathcal{A}_2$) holds. Then, $\exists \rho^* > 0$ such that for all $\rho \in (0, \rho^*)$ the following feedback control

$$v_u(k) = \begin{cases} -\rho \frac{\langle A_u x_u(k), B_u x_u(k-h) \rangle}{\|x_u^k\|_{\mathcal{C}_u}^2}, & \text{if } \|x_u^k\|_{\mathcal{C}_u} \neq 0 \\ 0, & \text{otherwise,} \end{cases} \quad (13)$$

exponentially stabilizes the system (1).

Idea of the Proof

Stabilization of subsystem (12) :

From the variation of constants formula and Gronwall's inequality, we obtain

$$\|x_s(k)\| \leq \alpha^{-h} e^{\ln(m(\rho))k} \|\psi_s\|_{\mathcal{C}_s}$$

for $\rho \in (0, \rho_s)$, $\ln(m(\rho)) < 0$.

Stabilization of subsystem (11) :

From **Theorem 3**, system (11) is uniformly exponentially stabilizable by the feedback control (13), for all $\rho \in (0, \rho_u)$, with $\rho_u := \frac{2}{\|B_u\|^2}$.

Conclusion:

Therefore, we deduce that for any $\rho \in (0, \rho^*)$, where $\rho^* := \min \{\rho_u, \rho_s\}$, the feedback control (13) uniformly exponentially stabilizes system (1).

Examples and simulations

Example 1

Consider the following delayed discrete semilinear system:

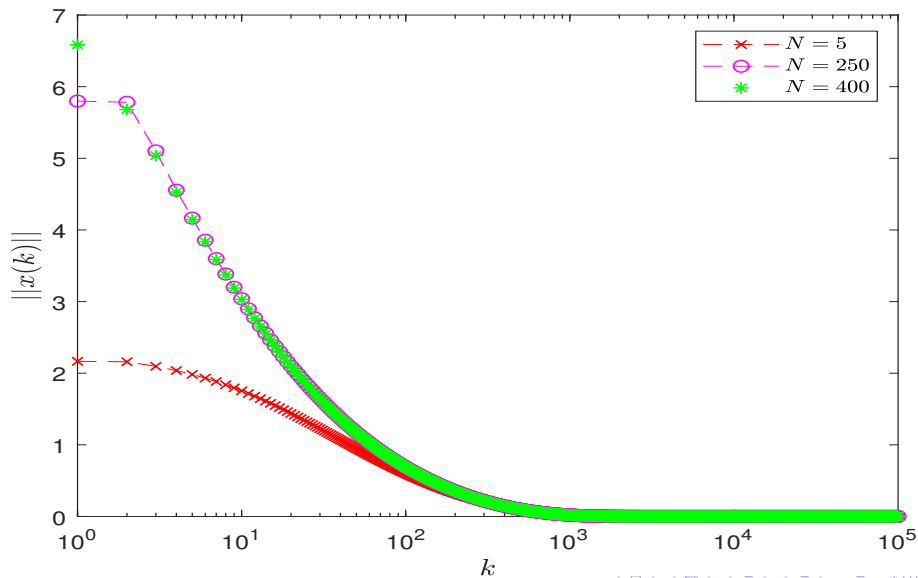
$$\begin{cases} x(k+1) = x(k) + v(k)B(x_k), & k \in \mathbb{N} \\ x_0 = \varphi \in \mathcal{C}, \end{cases} \quad (14)$$

where $\mathcal{H} = \ell^2(\mathbb{C})$ and $B(x_k) = \sum_{n=0}^{+\infty} x_n(k)(1 + g(x(k), x(k-h)))e_n$ such that $g : \mathcal{H} \times \mathcal{H} \longrightarrow \mathbb{C}$ is a locally Lipschitz function with $g(0, 0) = 0$. For a well-chosen function g satisfying

$$|1 + g(z_1, z_2)| \geq K, \quad \forall z_1, z_2 \in \mathcal{H},$$

where $K > 0$, $(\mathcal{A}_1)$ is verified for $r = 2$ with $\delta = K$.

Exponential Stabilization



Example 2

Consider the following delayed heat equation:

$$\begin{cases} \frac{\partial x}{\partial t}(t, \xi) = \Delta x(t, \xi) + v(t)B_c x(t-d, \xi), & (t, \xi) \in (0, +\infty) \times \Omega \\ \frac{\partial x}{\partial \eta}(t, \xi) = 0, & (t, \xi) \in (0, +\infty) \times \partial\Omega \\ x(t, \xi) = \psi(t, \xi), & (t, \xi) \in [-d, 0] \times \Omega, \end{cases}$$

where $\Omega = (0, 1)$ and $\mathcal{H} = L^2(\Omega)$, and $B_c y = \sum_{j=0}^{+\infty} \beta_j \langle y, \Phi_j \rangle_{L^2(\Omega)} \Phi_j$, where $(\beta_j)_{j \in \mathbb{N}}$ is a nonincreasing sequence of positive numbers.

- $(\Phi_j)_{j \in \mathbb{N}}$ is an orthonormal basis of $L^2(\Omega)$ defined by $\Phi_0(\xi) = 1$ and $\Phi_j(\xi) = \sqrt{2} \cos(j\pi\xi)$, $j \geq 1$, where $\xi \in \Omega$, composed of eigenfunctions with Neumann boundary conditions of the Laplacian operator with nonincreasing sequence of eigenvalues $(\lambda_j)_{j \in \mathbb{N}} := (-\pi^2 j^2)_{j \in \mathbb{N}}$.

By discretization and using the decomposition approach, we obtain:

$$\mathcal{H}_s := \text{span}(\varphi_j; 1 \leq j) \text{ and } \mathcal{H}_u := \text{span}(\varphi_0),$$

$$A = (I - rA_c)^{-1} \text{ and } B = r(I - rA_c)^{-1}B_c, \quad r \in (0, +\infty), \text{ with}$$

$$A_s y_s = \sum_{j=1}^{+\infty} (1 - r\lambda_j)^{-1} \langle y_s, \Phi_j \rangle \Phi_j, \quad A_u = I_{\mathcal{H}_u},$$

$$B_s y_s = r \sum_{j=1}^{+\infty} \beta_j (1 - r\lambda_j)^{-1} \langle y_s, \Phi_j \rangle \Phi_j \text{ and } B_u = r\beta_0 I_{\mathcal{H}_u}, \text{ with}$$

$$\|A_s\| = \sup\left(\frac{1}{1-r\lambda_j}\right) = \frac{1}{1+r\pi^2} < 1 \text{ and } \|A_u\| = 1.$$

For $y_u \in \mathcal{H}_u$, we have

$$\langle B_u A_u^{k-h} y_u, A_u^{k+1} y_u \rangle = \langle B_u y_u, y_u \rangle = r \beta_0 y_u^2, \quad \forall k \geq h.$$

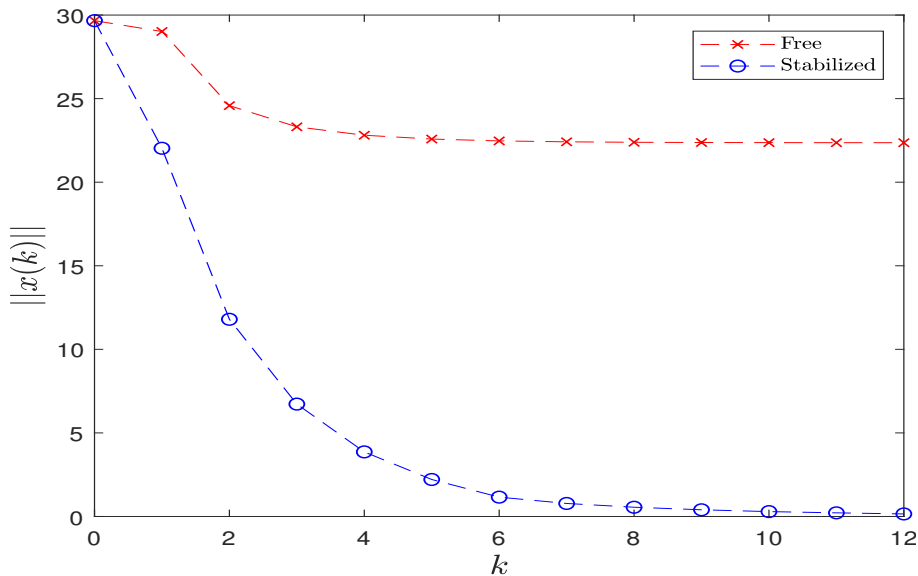
Then $\langle B_u A_u^{k-h} y_u, A_u^{k+1} y_u \rangle = 0 \Rightarrow y_u = 0$.

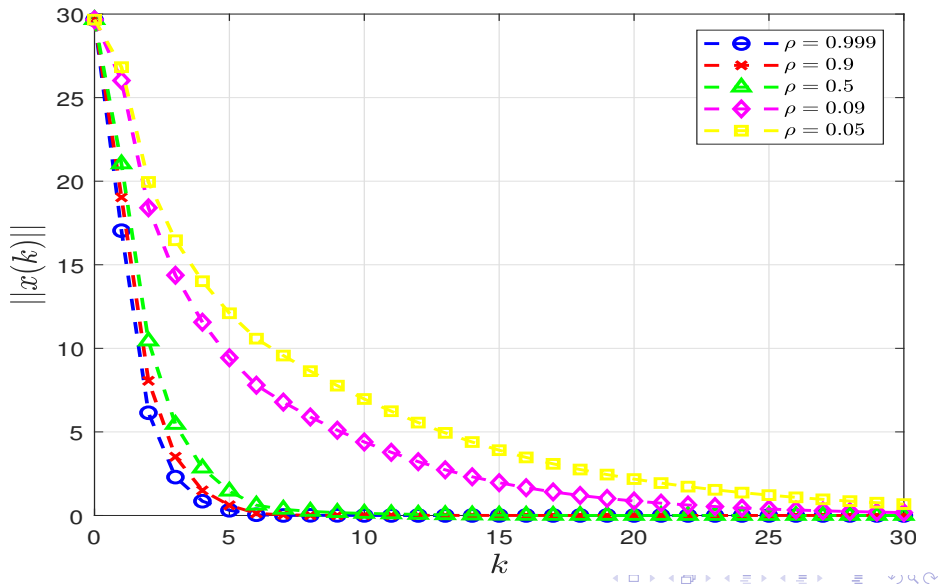
Hence, $(\mathcal{A}_2)$ is satisfied. According to **Theorem 3**, the following feedback control uniformly exponentially stabilizes the global system

$$v_u(k) = \begin{cases} -\frac{r \rho x_{1,u}(k) x_{1,u}(k-h)}{\|x_{1,u}^k\|_{\mathcal{C}_u}^2}, & \text{if } \|x_{1,u}^k\|_{\mathcal{C}_u} \neq 0 \\ 0, & \text{otherwise} \end{cases} \quad (15)$$

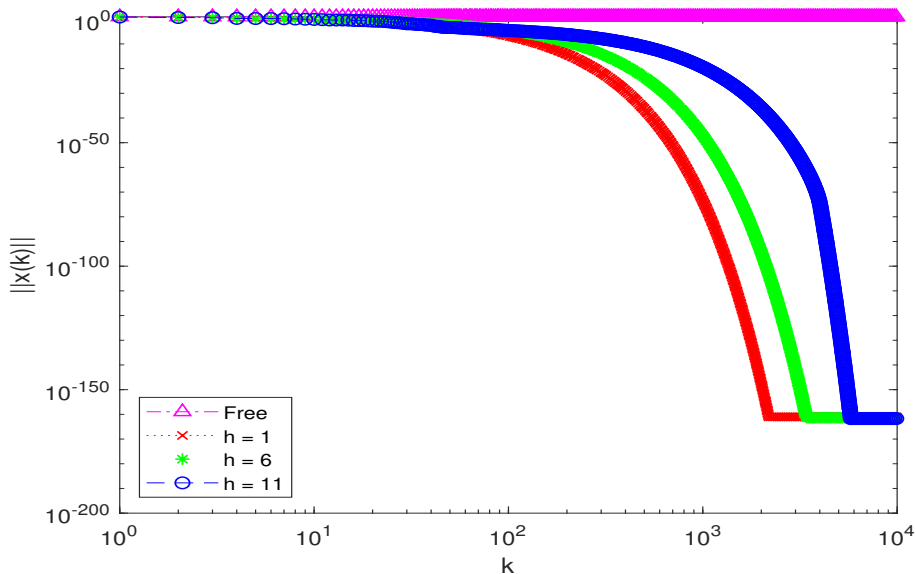
for all $\rho \in (0, \rho^*)$ with $\rho^* := \min \left\{ 1, \frac{2}{r^2}, \frac{\pi^2}{r(1+r\pi^2)^h} \right\}$.

The evolution of energy of the global system



Effect of ρ 

Effect of delay h



Conclusion and future directions

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- Future directions

- What if $\|A\| > 1$?

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For $r = 2$, exponential stabilization and for $r > 2$, polynomial stabilization have been achieved.

- Bilinear case has been discussed using a decomposition approach.

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- What if $\|A\| > 1$?
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Conclusion and Future directions

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- What if $\|A\| > 1$?
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Thank you for your attention!